

Ritesh Thawkar

AI Researcher · Computer Vision · Multimodal Systems

Abu Dhabi, UAE | riteshthawkar2003@gmail.com | +91 9552361592

[GitHub](#) | [Google Scholar](#) | [Hugging Face](#) | [LinkedIn](#) | [Portfolio](#)

Profile

Researcher focused on LLMs/VLMs, agentic RAG, and multimodal systems, with experience deploying real-world LLM applications and conducting research in self-evolving LLMs/VLMs and unified understanding-generation models.

Education

MSc Computer Vision

MBZUAI

Aug 2024 - Present

GPA: 3.55 / 4.00

BTech Computer Science & Engineering

Dr. Babasaheb Ambedkar Technological University

Feb 2021 - May 2024

GPA: 8.51 / 10.00

Experience

Research Assistant

Mohamed Bin Zayed University of Artificial Intelligence

Sep 2024 - Present

Abu Dhabi, UAE

- Developing Novel Visual Reasoning Benchmarks to stress-test the sequential logic capabilities of VLLMs.
- Pioneering Composed Video Retrieval techniques utilizing dense video descriptions for fine-grained understanding.
- Engineered core document retrieval systems and scalable web agents powering the enterprise Lawa.AI platform.
- Built and optimized the AI infrastructure and retrieval backends for the Nutrigenics.care startup platform.

Framework Engineer Intern

Quantiphi Analytics

Feb 2024 - May 2024

Remote

- Architected robust data-ingestion pipelines tailored for tracking small to medium business operations.
- Developed ML-driven analytics systems to automate the extraction of actionable business insights.
- Optimized and streamlined enterprise data processing workflows, reducing overall processing time by 20%.

Publications

arXiv 2026 How Good are Foundation Models in Longitudinal MRI Disease Progression Reasoning?

Wafa Al Ghallabi, **Ritesh Thawkar**, Sara Ghaboura, Omkar Thawakar, Numan Saeed, Dana Al Nuaimi, Ajnas Alkatheeri, Salman Khan, Fahad Shahbaz Khan. [View Preprint](#).

arXiv 2026 Perception Before Supervision: Self-Contained Visual Distillation from Counterfactual

Blind Spots. Shravan Venkatraman, Omkar Thawakar, **Ritesh Thawkar**, Abdelrahman Shaker, Rao Muhammad Anwer. [View Preprint](#).

arXiv 2026 Ask, Solve, Generate: Self-Evolving Unified Multimodal Understanding and Generation via

Self-Consistency Rewards. **Ritesh Thawkar**, Shravan Venkatraman, Omkar Thawakar, Abdelrahman Shaker, Fahad Khan, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. [View Preprint](#).

arXiv 2026 Paying More Attention to Visual Tokens in Self-Evolving Large Multimodal Models.

Shravan Venkatraman, **Ritesh Thawkar**, Omkar Thawakar, Rao Muhammad Anwer, Hisham Cholakkal, Salman Khan, Fahad Khan. [View Preprint](#).

WACV 2026 A Multi-Agent Diffusion Approach for MRI Anomaly Segmentation via Modality-Specific

LoRA Specialization. Wafa Al Ghallabi, Muhammad Zaigham Zaheer, **Ritesh Thawkar**, Omkar Thawakar, Salman Khan, Fahad Shahbaz Khan. [View Paper](#).

arXiv 2026 Mobile-O: Unified Multimodal Understanding and Generation on Mobile Device.

Abdelrahman Shaker, Ahmed Heakl, Jaseel Muhammad, **Ritesh Thawkar**, Omkar Thawakar, Senmao Li, Hisham Cholakkal, Ian Reid, Eric P. Xing, Salman Khan, Fahad Shahbaz Khan. [View Preprint](#).

arXiv 2025 EvoLMM: Self-Evolving Large Multimodal Models with Continuous Rewards. Omkar Thawakar, Shravan Venkatraman, **Ritesh Thawkar**, Abdelrahman Shaker, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Fahad Khan. [View Preprint](#).

arXiv 2025 How Good are Foundation Models in Step-by-Step Embodied Reasoning? Dinura Dissanayake, Ahmed Heakl, Omkar Thawakar, Noor Ahsan, **Ritesh Thawkar**, Ketan More, Jean Lahoud, Rao Anwer, Hisham Cholakkal, Ivan Laptev, Fahad Shahbaz Khan, Salman Khan. [View Preprint](#).

ICCV 2025 Beyond Simple Edits: Composed Video Retrieval with Dense Modifications. Omkar Thawakar, Dmitry Demidov, **Ritesh Thawkar**, Rao Muhammad Anwer, Mubarak Shah, Fahad Shahbaz Khan, Salman Khan. [View Paper](#).

EMNLP 2025 Fann or Flop: A Multigenre, Multiera Benchmark for Arabic Poetry Understanding in LLMs. Wafa Al Ghallabi, **Ritesh Thawkar**, Sara Ghaboura, Ketan Pravin More, Omkar Thawakar, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. [View Paper](#).

arXiv 2025 DriveLMM-o1: A Step-by-Step Reasoning Dataset and Large Multimodal Model for Driving Scenario Understanding. Ayesha Ishaq, Jean Lahoud, Ketan More, Omkar Thawakar, **Ritesh Thawkar**, Dinura Dissanayake, Noor Ahsan, Yuhao Li, Fahad Shahbaz Khan, Hisham Cholakkal, Ivan Laptev, Rao Muhammad Anwer, Salman Khan. [View Preprint](#).

ACL Findings 2025 Time Travel: A Comprehensive Benchmark to Evaluate LMMs on Historical and Cultural Artifacts. Sara Ghaboura, Ketan Pravin More, **Ritesh Thawkar**, Wafa Al Ghallabi, Omkar Thawakar, Fahad Shahbaz Khan, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. [View Paper](#).

ACL Findings 2025 LlamaV-o1: Rethinking Step-by-step Visual Reasoning in LLMs. Omkar Thawakar, Dinura Dissanayake, Ketan Pravin More, **Ritesh Thawkar**, et al. [View Paper](#).

Selected Projects

MBZUAI Web Agent Suite

Production

Agentic RAG, Web Agents, LLM Orchestration, FastAPI

Production web-agent system answering MBZUAI website queries with retrieval, tool use, and response verification.

Lawa Agent Studio

Private Platform

LLM, RAG, Workflow Orchestration, React, FastAPI

Platform for creating customizable LLM-powered agentic chatbots with configurable tools, tone, and knowledge sources.

Lawa.ai

Live Product

LLM, RAG, Workflow Orchestration, FastAPI, React

Enterprise agentic AI platform for multilingual, privacy-first assistants and revenue-generating deployments. [Visit Lawa.ai](#)

Nutrigenics.care

Startup Platform

LLM, Django, PostgreSQL, Recommendation System

Personalized nutrition platform with an image-aware Nutrition-GPT engine and recommendation workflows. [Visit Nutrigenics.care](#)

Technical Skills

AI Systems: Agentic RAG, VLMs / MLLMs, LLM Evaluation, Vector Databases

Languages: Python, TypeScript, JavaScript, C

ML / Deep Learning: PyTorch, Transformers, TensorFlow, Scikit-learn, NumPy, Pandas

Web Technologies: FastAPI, Flask, Django, React, Tailwind CSS

Databases: PostgreSQL, MySQL, SQLite

Cloud & DevOps: AWS, Azure, Git